Online Resource 1

Supplementary material for "Ranked Set Sampling under Imperfect Ranking: An Efficiency Ordering of Ranking Mechanisms and a Cost-Aware Adaptive Design"

Statistical Papers

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This resource contains the proofs of Theorems **??** and **??** of the paper (Sections S1 and S2), a normal approximation to the decision probability of the two-stage design (Section S3), the fixed-count reference experiment (Section S4), and the sensitivity of the regret to pilot size and budget (Section S5). Theorem, equation, table, and section numbers without the prefix "S" refer to the paper.

S1 Proof of Theorem ?? (Huber M-estimator)

Step 1 (the population score). For all x and t, $\psi_c(x-t)-\psi_c(x-\theta) = -\int_\theta^t \mathbf{1}\{|x-u|<c\}\,du$, because ψ_c is absolutely continuous with derivative $\mathbf{1}\{|u|<c\}$ almost everywhere. By Fubini's theorem,

$$\lambda_F(t) = -\int_\theta^t \pi(u)\,du, \qquad \pi(u) = \Pr_F(|X-u|<c).$$

By (A1), π is continuous at $u=\theta$ with $\pi(\theta)=A_F$. Hence λ_F is non-increasing and differentiable at θ with $\lambda_F'(\theta) = -A_F < 0$, so that θ is its unique zero and $\lambda_F(\theta + x/\sqrt{n}) = -A_F x/\sqrt{n} + o(n^{-1/2})$ for every fixed x.

Step 2 (monotonicity). The function $G_n(t) = \sum_{i=1}^n \psi_c(X_i - t)$ is continuous and non-increasing, with $G_n(t) = nc$ for small t and $G_n(t) = -nc$ for large t, so a solution exists and, for every t,

$$\{G_n(t) < 0\} \subseteq \{\widehat{\theta}_n \le t\} \subseteq \{G_n(t) \le 0\}.$$

Step 3 (mean and variance of the score sum). Fix x and put $t_n = \theta + x/\sqrt{n}$. The n observations are independent, with exactly k of them drawn from each g_s. By Eq. (**??**),

$$E[G_n(t_n)] = k\sum_{s=1}^m E_{g_s}[\psi_c(X-t_n)] = n\lambda_F(t_n) = -\sqrt{n}\,A_F x + o(\sqrt{n}).$$

Since $|\psi_c(x-t_n)-\psi_c(x-\theta)| \le |t_n-\theta|$ uniformly in x and $|\psi_c|\le c$,

$$s_n^2 := \mathrm{Var}\{G_n(t_n)\} = k\sum_{s=1}^m \mathrm{Var}_{g_s}\{\psi_c(X-t_n)\} = n\{\sigma_\psi^2 + o(1)\}.$$

Step 4 (central limit theorem). The centered summands of $G_n(t_n)$ are independent and bounded by $2c$, and $s_n \to \infty$ by (A2); the Lindeberg condition therefore holds and $\{G_n(t_n) - E[G_n(t_n)]\}/s_n \to N(0,1)$ in distribution. Consequently

$$\Pr\{G_n(t_n)\le 0\} = \Pr\left\{\frac{G_n(t_n)-E[G_n(t_n)]}{s_n} \le \frac{\sqrt{n}\,A_F x + o(\sqrt{n})}{s_n}\right\} \longrightarrow \Phi\left(\frac{A_F x}{\sigma_\psi}\right),$$

where the central limit theorem is the Lindeberg–Feller theorem for the triangular array of summands $\psi_c(X_i - t_n)$, and the convergence of distribution functions to the continuous limit Φ is uniform (Pólya's theorem), which permits the moving threshold; the same limit holds for $\Pr\{G_n(t_n) < 0\}$ because the limit law is continuous. By Step 2, $\Pr\{\sqrt{n}(\widehat{\theta}_n - \theta) \le x\} \to \Phi(A_F x/\sigma_\psi)$ for every x, which is the stated normal limit.

Step 5 (variance ordering). By definition, $V_{\mathrm{IF,RSS}} = m^{-1}\sum_s \mathrm{Var}_{g_s}\{\psi_c(X-\theta)/A_F\} = \sigma_\psi^2/A_F^2$. The inequality and its equality condition are the general influence-function comparison displayed before the theorem, applied to the bounded function $\psi_c(\cdot - \theta)/A_F$, whose F-mean is zero. □

S2 Proof of Theorem ?? (trimmed mean)

Let G_s be the distribution function of g_s and $G_{s,k}$ the empirical distribution function of the k measured units of observed rank s, so that $F_n = m^{-1}\sum_s G_{s,k}$ and $F = m^{-1}\sum_s G_s$. The $G_{s,k}$ are independent, and by Donsker's theorem $\sqrt{k}(G_{s,k} - G_s)$ converges weakly in $\ell^\infty(\mathbb{R})$ to $\mathbb{B}_s \circ G_s$, with $\mathbb{B}_1, \dots, \mathbb{B}_m$ independent Brownian bridges. Hence

$$\sqrt{n}\,(F_n - F) = m^{-1/2}\sum_{s=1}^{m}\sqrt{k}\,(G_{s,k} - G_s) \;\rightsquigarrow\; \mathbb{G} = m^{-1/2}\sum_{s=1}^{m}\mathbb{B}_s \circ G_s.$$

Because $G_s \le mF$ as measures, each G_s is continuous wherever F is, so $\mathbb{G}$ has continuous sample paths on the interval of the assumption. Under that assumption the quantile transformation $F \mapsto F^{-1}$, viewed as a map into $\ell^\infty[\alpha, 1-\alpha]$, is Hadamard differentiable at F tangentially to the continuous functions, with derivative $\eta \mapsto -(\eta/f)\circ F^{-1}$ (van der Vaart, 1998, Lemma 21.4). Integration over $[\alpha, 1-\alpha]$ is linear and continuous, so T_α is Hadamard differentiable with derivative

$$T_F'(\eta) = -\frac{1}{1-2\alpha}\int_\alpha^{1-\alpha}\frac{\eta(F^{-1}(u))}{f(F^{-1}(u))}\,du = -\frac{1}{1-2\alpha}\int_{q_\alpha}^{q_{1-\alpha}}\eta(x)\,dx.$$

The functional delta method (van der Vaart, 1998, Theorem 20.8) gives $\sqrt{n}\{T_\alpha(F_n) - T_\alpha(F)\} \rightsquigarrow T_F'(\mathbb{G})$, a centered normal variable. For a point mass, $T_F'(\mathbf{1}[y,\infty) - G) = \{\mathcal{W}(y) - E_G[\mathcal{W}]\}/(1-2\alpha)$ for any distribution function G, so $T_F'(\mathbb{B}_s \circ G_s)$ is normal with variance $\mathrm{Var}_{g_s}\{\mathrm{IF}(X; T_\alpha, F)\}$, and by independence the limit variance is $m^{-1}\sum_s \mathrm{Var}_{g_s}(\mathrm{IF}) = V_{\mathrm{IF,RSS}}$. The inequality and its equality case are the influence-function comparison of this section applied to the bounded function $\mathrm{IF}(\cdot; T_\alpha, F)$, which has F-mean zero. The integer-trimmed sample mean and $T_\alpha(F_n)$ differ only in the weights of at most two order statistics at each boundary, by $O(n^{-1})$ each, and these order statistics are bounded in probability. □

S3 Normal approximation to the decision probability

Proposition S1. *Under the assumptions of Proposition* **??**, *suppose that* $E_F[h^4] < \infty$ *and* $\mathrm{RE}_h(P) > 1$. *Then, as* $k_0 \to \infty$ *with* m *fixed,*

$$\sqrt{k_0}\left\{\widehat{\mathrm{RE}}_h - \mathrm{RE}_h(P)\right\} \xrightarrow{d} N(0, s_h^2), \qquad s_h^2 = \frac{1}{m^2\bar{w}^2}\sum_{s=1}^{m}\Big\{4d_s^2\sigma_s^2 - 4\varrho\, d_s\mu_{3,s} + \varrho^2(\mu_{4,s} - \sigma_s^4)\Big\}, \tag{S1}$$

where σ_s^2, $\mu_{3,s}$, $\mu_{4,s}$ *are the central moments of* $h(X)$ *under* g_s, $d_s = E_{g_s}[h] - E_F[h]$, $\bar{w} = m^{-1}\sum_s \sigma_s^2$, *and* $\varrho = \mathrm{RE}_h(P) - 1$. *Consequently, for a threshold* τ,

$$\Pr\left(\widehat{\mathrm{RE}}_h > \tau\right) \approx \Phi\left(\frac{\sqrt{k_0}\,\{\mathrm{RE}_h(P) - \tau\}}{s_h}\right). \tag{S2}$$

Proof. With the notation of the proof of Proposition **??**, the central limit theorem in each stratum gives $\sqrt{k_0}(\bar{h}_s - \mu_{h,s},\, S_s^2 - \sigma_s^2) \to N(0, \Sigma_s)$ with Σ_s having entries σ_s^2, $\mu_{3,s}$, $\mu_{4,s} - \sigma_s^4$, independently over s. Writing $\varepsilon_s = \bar{h}_s - \mu_{h,s}$ and using $\sum_s d_s = 0$, $m^{-1}\sum_s(\bar{h}_s - \bar{h})^2 = \bar{b} + 2m^{-1}\sum_s d_s\varepsilon_s + O_p(k_0^{-1})$ with $\bar{b} = m^{-1}\sum_s d_s^2$, so $\sqrt{k_0}(B - \bar{b}) = 2m^{-1}\sum_s d_s\sqrt{k_0}\,\varepsilon_s + o_p(1)$, while $\sqrt{k_0}(W - \bar{w}) = m^{-1}\sum_s \sqrt{k_0}(S_s^2 - \sigma_s^2)$. The delta method applied to $(B, W) \mapsto 1 + B/W$, with gradient $(1/\bar{w}, -\varrho/\bar{w})$ at $(\bar{b}, \bar{w})$, gives the stated variance; the truncation at one is asymptotically inactive because $\mathrm{RE}_h(P) > 1$. □

Eq. (S2) with $\tau = 1 + \gamma$ approximates $p_R = \Pr(D = \mathrm{R})$ for large pilots. It is not valid at $\mathrm{RE}_h(P) = 1$, where B is of order k_0^{-1}. In the simulations of Section **??** it is adequate near the break-even point for $k_0 \geq 20$ but poor for small pilots and near $\mathrm{RE} = 1$ (0.005 against a simulated 0.125 at $\rho_R = 0.25$ with $k_0 = 5$); the noncentral-F approximation of Eq. (**??**) is used in the paper for this reason.

S4 Fixed-count reference experiment

This experiment compares RSS and SRS at the same number of measured observations, without a ranking cost.

S4.1 Design

The contamination experiments distinguish two scientifically different placements for RSS. Under *post-ranking* contamination, ranking is performed on the uncontaminated responses and contamination is introduced afterward, so the ranking mechanism itself is unchanged. Under *pre-ranking* contamination, the contaminated responses are present during ranking, so contamination can alter which units occupy each true rank and therefore which units enter the measured sample. In the implemented mechanism $V_i = R_i + \nu Z_i$ the true rank is computed from the contaminated responses, so the ranker is an oracle on those values up to rank noise; the rank-only model then holds exactly with F_ϵ, but this is not a model of a ranker misled by contamination. The fixed-strategy reference experiment uses $\epsilon \in \{0, 0.10\}$ under an exact replacement contamination model: exactly $\lfloor \epsilon N \rfloor$ of the $N = km^2$ candidate responses are replaced by 8ξ with $\xi \in \{-1, +1\}$ sampled independently with probability 1/2 each. For SRS, the same contaminated candidate pool is formed and an SRS of size $n = km$ is then drawn without replacement; consequently the number of contaminated SRS observations is random (hypergeometric), rather than fixed at $\lfloor \epsilon n \rfloor$. The pre/post distinction has no ranking-stage effect for SRS. For pre-ranking contamination, the displayed ρ_R remains the clean-calibrated target for the ranking mechanism $V_i = R_i + \nu Z_i$; contamination changes the candidate response values used in the ranking step but does not alter this rank-noise mechanism itself. Thus, the displayed target is a clean-mechanism calibration, and comparisons between pre- and post-ranking conditions at the same displayed

target should not be interpreted as comparisons of identical contaminated-value distributions. In the balanced rank-only model, the averaging identity also implies that the measured sample has contamination fraction ϵ in expectation; preferential selection of contaminated observations, in the sense of changing that expected fraction, requires unbalanced allocation or value-dependent ranking and is outside the present model. Ties in the contaminated values are resolved by the stable input-order rule used by the implementation. Under exact replacement, the contamination indicators are exchangeable but dependent because their total number is fixed. If contamination were instead i.i.d. with law H, the same rank-only derivation would apply exactly with F replaced by $F_\epsilon = (1-\epsilon)F + \epsilon H$ for pre-ranking contamination; for post-ranking contamination, each selected stratum would have law $(1-\epsilon)g_s + \epsilon H$. In the present fixed-count implementation these formulas provide the corresponding marginal/first-moment approximation, while the fixed total induces cross-set dependence, so the clean finite-sample variance identity is not claimed as exact for the contaminated experiment.

Values above one indicate lower fixed-strategy variance for RSS, values below one indicate lower variance for SRS, and values near one indicate little practical difference. The reference experiment also computes MSE as a consistency check. Under the present symmetric $N(0,1)$ model with equiprobable ± 8 replacement contamination, the contamination positions are sampled independently of the response values. The mean and symmetric trimmed mean are odd-equivariant, and the Huber location estimator is odd-equivariant with a symmetric choice among minimizers when the minimizer is non-unique. The Gaussian and Laplace ranking-noise laws are symmetric, the balanced permutation design is invariant under rank reversal, and the resulting ranking mechanisms satisfy the reflection symmetry $P_{m+1-r,m+1-s} = P_{rs}$. Stable tie-breaking does not break this symmetry because tied values are identical. Hence the RSS and SRS sampling distributions are symmetric about zero whenever the estimator expectations exist, so these estimators have zero finite-sample bias under the specified design and MSE and variance coincide. Empirically, the two reference criteria identify the same oracle design in every condition; no separate MSE-efficiency claim is made. Monte Carlo uncertainty is reported using bootstrap intervals (Efron, 1979). Breakdown is not inferred from the finite grid of contamination percentages; a breakdown claim requires a separate contamination experiment that searches for divergence or instability under the stated contamination model.

Each condition draws responses from the standard normal model $X \sim N(0,1)$ before contamination, uses $m = 5$ sets per cycle, 100 cycles, and hence $n = 500$ measured observations. The clean-calibrated ranking targets are $\rho_R \in \{0.25, 0.50, 0.75, 0.90\}$; the ranking-error mechanism is Gaussian or Laplace; contamination is absent or has fraction $\epsilon = 0.10$; and contamination is placed either before or after ranking. The three estimators are the sample mean, 10% trimmed mean, and Huber M-estimator. This produces 96 labeled experimental conditions. Because the two contamination-placement labels are distributionally identical when $\epsilon = 0$, the factorial grid contains 72 distinct data-generating configurations plus 24 clean configurations duplicated under the two placement labels; the reported aggregate summaries retain the planned 96-condition weighting.

The ranking scales are calibrated using 20,000 trials per target–mechanism pair. Table S1 reports the calibrated scales and the Monte Carlo calibration estimates; the residual is that

of the root-finder on fixed calibration draws; the Monte Carlo standard error of a calibration mean is about 0.003 in the weakest-rank case, so the displayed root-finding residual should not be interpreted as a statistical accuracy bound.

Table S1: Ranking-scale calibration at $m = 5$ (20,000 calibration trials per row).

Mechanism	Target ρ_R	Scale ν	Calibration-sample mean ρ_R	|Root-finding residual|
Gaussian	0.25	5.56036	0.24999	1.0×10^{-5}
Laplace	0.25	4.76666	0.25000	$< 10^{-12}$
Gaussian	0.50	2.50672	0.50000	$< 10^{-12}$
Laplace	0.50	2.07187	0.50000	$< 10^{-12}$
Gaussian	0.75	1.33957	0.75000	$< 10^{-12}$
Laplace	0.75	1.03538	0.75000	$< 10^{-12}$
Gaussian	0.90	0.795978	0.90000	$< 10^{-12}$
Laplace	0.90	0.579369	0.900005	5.0×10^{-6}

The fixed-strategy risks are estimated from 5,000 independent replications per condition (Table S2).

S4.2 Fixed-strategy reference risks

Define the fixed-strategy variance efficiency by

$$\mathrm{RE}_{V,\mathrm{ref}} = \frac{\widehat{V}^{\mathrm{ref}}_{SRS}}{\widehat{V}^{\mathrm{ref}}_{RSS}},$$

where $\widehat{V}^{\mathrm{ref}}_{SRS}$ and $\widehat{V}^{\mathrm{ref}}_{RSS}$ denote the sample variances of the SRS and RSS estimates, respectively, in the independent 5,000-replicate fixed-strategy reference experiment. Values above one indicate that RSS has lower fixed-strategy risk, while values below one indicate the opposite. Averaged over the 24 conditions at each target ρ_R, the mean reference variance-efficiency ratios increase from 1.028 at $\rho_R = 0.25$ to 1.170, 1.477, and 1.884 at $\rho_R = 0.50, 0.75, 0.90$, respectively. The reference efficiency remains heterogeneous across mechanism, contamination placement, and estimator. These empirical results should not be interpreted as a universal efficiency ordering across the full contaminated simulation grid.

Table S2: Mean fixed-strategy variance efficiency by clean-calibrated target ranking quality (5,000 reference replications per condition).

Target ρ_R	Mean $\mathrm{RE}_{V,\mathrm{ref}}$
0.25	1.028
0.50	1.170
0.75	1.477
0.90	1.884

Across all 96 labeled conditions, the empirical fixed-strategy oracle is RSS in 90 conditions and SRS in 6 conditions, for both variance and MSE. The labels are defined by the smaller

5,000-replicate reference estimate and therefore should not be interpreted as population-level decisions. Five SRS-oracle labels occur at $\rho_R = 0.25$ and one at $\rho_R = 0.50$. All six SRS-oracle labels are contaminated conditions; five are post-ranking and one is pre-ranking, and four use the sample mean. Their condition-level 95% bootstrap intervals all contain one, with lower endpoints ranging from 0.912 to 0.942 and upper endpoints from 1.015 to 1.052. No condition among the 96 has a reference interval lying entirely below one, and six oracle labels changed between a preliminary 2,000-replicate reference run and the final 5,000-replicate run. These labels are therefore not distinguishable from RE = 1 at the present Monte Carlo precision and should not be treated as population-level evidence that SRS is intrinsically more efficient.

To make the interaction between contamination placement and estimator choice explicit in one representative contaminated setting, Table S3 reports the final 5,000-replicate reference relative efficiencies at $\rho_R = 0.90$ and $\epsilon = 0.10$, averaged across the Gaussian and Laplace ranking mechanisms. The values are computed from the final reference file and are therefore a deterministic post-processing result, not an additional simulation run. Post-ranking contamination leaves the mean only slightly above the SRS benchmark (RE = 1.050), whereas the 10% trimmed mean and Huber estimator retain substantially larger RSS gains (1.572 and 1.589, respectively). Under pre-ranking contamination, the corresponding efficiencies are 1.439, 2.117, and 2.120. Thus, within this specified setting, the effect of contamination placement depends materially on the estimator used; because pre-ranking ranks the contaminated values themselves, the larger pre-ranking efficiencies also reflect that the ranker can see the outliers.

Table S3: Placement-by-estimator interaction at $\rho_R = 0.90$ and $\epsilon = 0.10$, using the final 5,000-reference-replicate experiment. Relative efficiency is $\mathrm{RE}_{V,\mathrm{ref}} = \widehat{V}^{\mathrm{ref}}_{SRS}/\widehat{V}^{\mathrm{ref}}_{RSS}$; Gaussian and Laplace columns are mechanism-specific, and the final column averages the two mechanisms.

Placement	Estimator	Gaussian RE	Laplace RE	Mean RE
Post-ranking	Huber	1.613536	1.563751	1.588644
Post-ranking	Mean	1.034148	1.065911	1.050029
Post-ranking	10% trimmed	1.596496	1.548292	1.572394
Pre-ranking	Huber	2.102955	2.137148	2.120051
Pre-ranking	Mean	1.430783	1.448025	1.439404
Pre-ranking	10% trimmed	2.093944	2.139074	2.116509

S4.3 Monte Carlo uncertainty and the cost-free case

Across the 96 conditions the mean fixed-strategy efficiency is 1.390 (95% bootstrap interval $[1.382, 1.398]$) and the median is 1.225 ($[1.213, 1.259]$). Bootstrap resampling is performed within each condition over its 5,000 reference replicates, with paired resampling for each RSS–SRS ratio, and the condition-level estimates are then aggregated. Perfect-ranking and random-ranking controls were not part of this reference experiment; they are included in the cost-aware experiment of Section **??**.

An earlier version of this study examined a cost-free selection rule that chose between RSS and SRS by comparing variances estimated from 200 separate pilot replications of 20 cycles

each. On the same 96 conditions it selected RSS in 83.97% of 1,000 trials per condition, and its mean variance regret was 1.08% (2.82% in the realized trials) against 0.11% for always using RSS. This is the behaviour predicted by Remark **??**(d): without a ranking cost the efficiency is at least one, SRS selections are driven by pilot noise, and switching cannot help. That rule is therefore not pursued, and the adaptive design studied below is the cost-aware design of Section **??**.

S4.4 Comparison with the first-order predictions

The first-order contaminated-law calculation was also evaluated for all three estimators used in the primary study. Table S4 reports the mechanism-averaged predictions from the corresponding transition-matrix calculation alongside the final 5,000-replicate reference efficiencies. The comparison is intended as a theory–simulation alignment check, not as an exact finite-sample validation theorem for the fixed-count contamination protocol.

Table S4: Theory–simulation comparison for clean and contaminated conditions, averaged over Gaussian and Laplace ranking mechanisms. Clean predictions are the high-precision mean-efficiency calculations in Table **??**; contaminated predictions use the i.i.d. contaminated-law approximation. Observed values come from the final 5,000-replicate fixed-strategy reference experiment. Bracketed values are condition-aggregated nonparametric bootstrap 95% intervals.

ρ_R	Placement	Estimator	Predicted RE	Observed RE (95% CI)
0.25	Clean	Mean	1.041	1.051 [1.023,1.080]
0.50	Clean	Mean	1.190	1.200 [1.167,1.232]
0.75	Clean	Mean	1.556	1.545 [1.504,1.589]
0.90	Clean	Mean	2.059	2.048 [1.995,2.103]
0.25	Post	Mean	1.004	0.978 [0.941,1.017]
0.25	Post	10% trimmed	1.028	1.004 [0.966,1.044]
0.25	Post	Huber	1.030	1.004 [0.966,1.043]
0.25	Pre	Mean	1.026	0.992 [0.955,1.031]
0.25	Pre	10% trimmed	1.043	1.018 [0.980,1.058]
0.25	Pre	Huber	1.043	1.019 [0.980,1.058]
0.50	Post	Mean	1.018	1.012 [0.973,1.052]
0.50	Post	10% trimmed	1.124	1.122 [1.081,1.166]
0.50	Post	Huber	1.132	1.126 [1.083,1.169]
0.50	Pre	Mean	1.113	1.139 [1.096,1.184]
0.50	Pre	10% trimmed	1.199	1.202 [1.157,1.249]
0.50	Pre	Huber	1.199	1.203 [1.159,1.250]
0.75	Post	Mean	1.041	1.049 [1.010,1.089]
0.75	Post	10% trimmed	1.331	1.358 [1.307,1.408]
0.75	Post	Huber	1.356	1.369 [1.319,1.421]
0.75	Pre	Mean	1.293	1.308 [1.259,1.359]
0.75	Pre	10% trimmed	1.595	1.611 [1.552,1.673]
0.75	Pre	Huber	1.597	1.608 [1.549,1.672]
0.90	Post	Mean	1.061	1.050 [1.011,1.093]
0.90	Post	10% trimmed	1.556	1.572 [1.514,1.634]
0.90	Post	Huber	1.607	1.589 [1.530,1.648]
0.90	Pre	Mean	1.488	1.439 [1.385,1.495]
0.90	Pre	10% trimmed	2.157	2.117 [2.035,2.200]
0.90	Pre	Huber	2.163	2.120 [2.043,2.204]

For these 28 evaluated conditions, every prediction lies inside the 95% interval of the corresponding observed efficiency. The intervals are wide (half-width 0.04 to 0.08), so

this comparison supports agreement only to within a few percent. The deviations are not independent across estimators if these are computed from the same replicates, and within a ranking-quality level they tend to share a sign: all six contaminated values at $\rho_R = 0.25$ fall 0.024 to 0.034 below their predictions, all six at $\rho_R = 0.75$ lie 0.008 to 0.027 above, and the three pre-ranking values at $\rho_R = 0.90$ lie 0.04 to 0.05 below. This experiment also uses fixed-count contamination, for which the F_ϵ calculation is not exact, so it cannot separate linearization error, fixed-count dependence, and Monte Carlo noise. The experiment with i.i.d. contamination reported in Table **??** of the paper separates these effects.

S5 Sensitivity to pilot size and budget

Table S5 reports the sensitivity to the pilot size and to the budget. The regret of the adaptive design is governed by the pilot overhead ω: it grows with the share of the budget spent on the pilot and decreases as the budget increases. A pilot of 10 cycles gave the smallest worst-case regret for $C = 500$; smaller pilots make more decision errors and larger pilots cost more than they save. The predicted worst-case regrets from Eqs. (**??**) and (**??**) for $k_0 = 5$, 10, 20, and 40 are 4.5%, 3.1%, 5.3%, and 11.1%, in agreement with the simulated 5.6%, 3.7%, 5.4%, and 10.6%, and the predicted minimum is at $k_0 = 11$ (Table **??**).

Table S5: Sensitivity of the regret (%) to pilot size and budget: clean mean, $\gamma = 0.25$, worst case and mean over the 10 ranking mechanisms (40,000 replications each; fixed-design regrets are simulated values relative to the exact optimum).

		Worst-case regret			Mean regret		
Budget C	Pilot k_0	SRS	RSS	Adaptive	SRS	RSS	Adaptive
500	5	121.3	25.4	5.6	30.0	7.9	2.9
500	10	120.2	26.4	3.7	29.6	7.3	1.3
500	20	122.0	24.3	5.4	30.2	7.4	1.8
500	40	123.1	25.2	10.6	30.4	7.4	3.7
250	20	122.0	26.0	10.7	30.1	7.6	3.8
1000	20	122.8	25.6	3.0	30.7	7.6	1.3

References

Efron B (1979) Bootstrap methods: another look at the jackknife. Ann Stat 7(1):1–26

van der Vaart AW (1998) Asymptotic statistics. Cambridge University Press, Cambridge